Misperceived Social Norms and Willingness to Act Against Climate Change

Based on Andre, Boneva, Chopra and Falk (2024)

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Outline

Humanity's Greatest Threat

- **Global Warming**[1]
  - **10–25%** Global GDP losses by 2100 [Undiminished] (SSP3-7.0 scenario)
  - **0.5–5%** Global GDP losses by 2100 [1.5–2°C] (Paris Agreement Target)

    Stated by the Intergovernmental Panel on Climate Change (IPCC) 2023.

- **Climate protection makes economic sense:**
  E.g. For every 1€ invested in climate mitigation, 4€ in avoided climate damages

A classic problem of free riding

The benefits of climate protection are globally distributed, but protecting the climate comes at a personal cost. How can we fight global warming?

[1] 'Global warming' is used to avoid confusion with short-term or seasonal weather changes, or ozone depletion Lorenzoni et al. 2006.

Research Questions

Andre et al. (2024b) provides the first comprehensive picture of the US climate context using a representative sample and an incentivised donation decision.

1. What factors drive individual willingness to fight Global Warming? **Behavioural Determinants**
2. Do misperceptions of social norms influence climate action, and if so, how? **Social Norms**
3. Can correcting misperceived norms increase support for climate policies? **Intervention Effect**

How to measure Individuals' Willingness to fight Global Warming

- Quota-Based sampling approach of 8,000 participants [2]
  - **Representing** the adult US population in terms of key socio-demographic variables, allowing for **mitigation of concerns about selection**.
  - Two separate collection waves with $n = 2000$ and $n = 6000$.
  - Respondents could only participate in one of the two waves.
- Collect detailed background information of individuals' moral values, preferences and beliefs about social norms.
- Study the **impact of the information treatment** and examine if the treatment effects of information are heterogeneous across all kinds of groups.

[2] Recrutement done by Pureprofile

Sample Characteristics

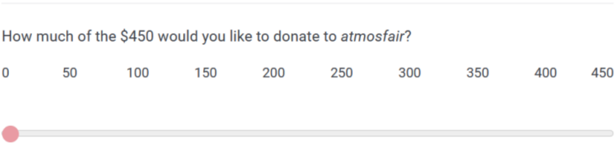
Table A.1: Comparison of the sample to the US population

Variable	Wave 1	Wave 2	US population (2019)
Female	51%	51%	51%
Age: 18-34	30%	30%	30%
Age: 35-54	32%	32%	32%
Age: 55+	38%	38%	38%
Education: Bachelor's degree or above	31%	31%	31%
Region: Northeast	17%	17%	17%
Region: Midwest	21%	21%	21%
Region: South	38%	38%	38%
Region: West	24%	24%	24%
Democrat*	54%	53%	52%
Republican*	46%	47%	47%
Income: Below 50k*	41%	45%	37%
Income: 50k to 100k*	39%	35%	31%
Income: Above 100k*	19%	20%	31%

Wave 1: Measuring Willingness to fight Global Warming

Descriptive Analysis (n = 2,000, March 2021)

- Participants divide $450 between themself & *atmosfair*[3].
- Before participants make their decision, the instructions provide further information on *atmosfair*.



- Results are quantitative & interpersonally comparable.
- Probabilistic incentive scheme: Decisions of 25 randomised participants are implemented.

[3] atmosfair is actively engaged in climate protection, including offsetting greenhouse gas emissions through renewable energy.

Wave 1: Measuring Behaviour Determinants

Descriptive Analysis (n = 2,000, March 2021)

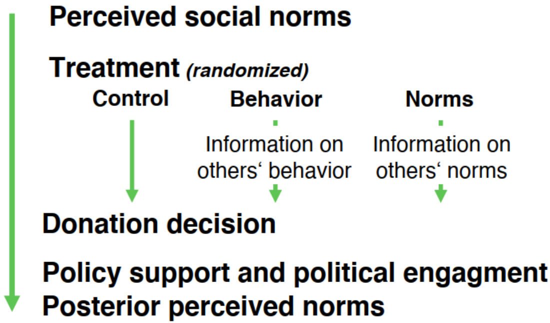
- Baseline behaviour and endorsements of norm [4]
  - Behaviour: 'Do you try to fight global warming?' [Yes/No]
  - Norms: 'Do you think that people in the United States should try to fight global warming?' [Yes/No]
- Perceived social norms
  - Perceived behaviour: 'Out of 100 people, how many stated that they try to fight global warming?' [0-100]
  - Perceived norms: 'Out of 100 people, how many stated that they think that people in the United States should try to fight global warming?' [0-100]
- Additional Measurements: Policy support, Political engagement, Posterior beliefs about social norms, Climate change scepticism, Background characteristics.

[4] The solution means are used as the Information treatments in Wave 2.

Wave 2: Shifting Perceived Social Norms

Information experiment (n = 6,000, April 2021)

Figure A.1: Structure of experiment



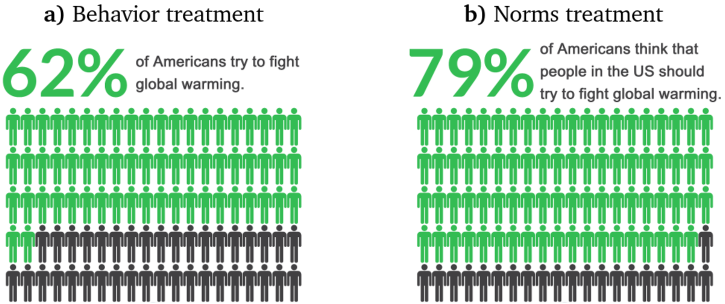
Notes: This figure provides an overview of the structure of the experiment.

Analysis Plan of Andre et al. (2021)

Wave 2: Information Treatments

Information experiment (n = 6,000, April 2021)

Figure B.5: Information Treatments

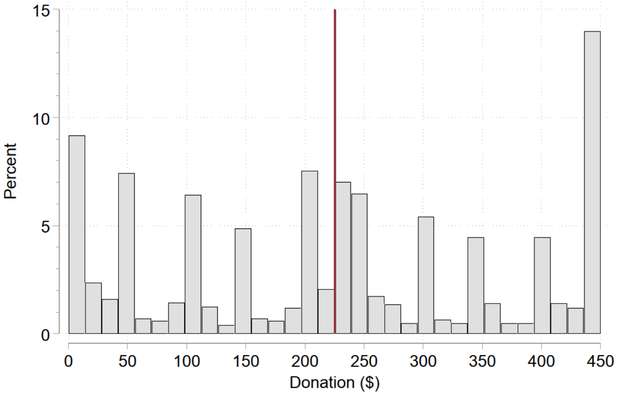


Results of the answers for $n = 2000$ Americans from Wave 1

Willingness to fight Global Warming and its Determinants

Results Wave 1

Figure A.2: The distribution of individual willingness to fight global warming



Substantial heterogeneity → Strong variation in individual willingness to act → search for determinants

Willingness to fight Global warming and its Determinants

Results Wave 1

Table 1: Determinants of climate change behaviour

	Donation ($)	
Variable	(1)	(2)
Perceived social norms		
Behavior belief	12.067***	
	(3.146)	
Norms belief		14.372***
		(3.058)
Economic preferences		
Altruism	51.361***	51.819***
	(3.476)	(3.447)
Patience	15.139***	15.132***
	(3.095)	(3.086)
Risk	-1.527	-0.925
	(3.374)	(3.357)
Positive reciprocity	9.504***	7.819**
	(3.243)	(3.260)
Negative reciprocity	-3.469	-2.690
	(3.212)	(3.183)
Trust	1.061	0.816
	(3.231)	(3.201)
Moral foundations		
Relative universalism	23.496***	23.137***
	(3.295)	(3.283)

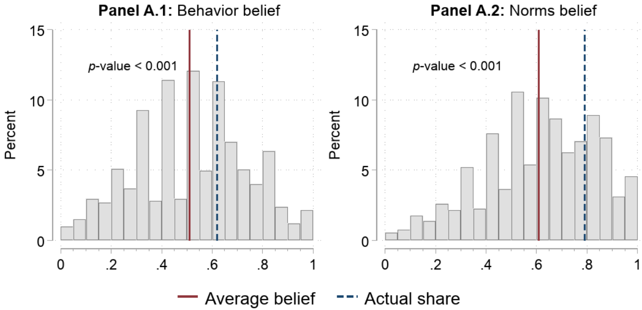
All predictors are z-scored ($z-score = \frac{X_i - \overline{X}}{SD}$)

Misperceived Social Norms

Results Wave 1

Figure 2: Perceived social norms: fight global warming

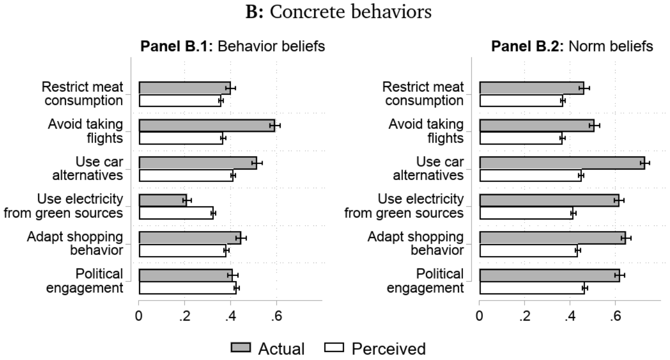
A: "Fight global warming"



Americans systematically underestimate the share of Americans **trying to fight** global warming as well as the share who **think that** Americans should **try to fight** global warming.

Misperceived Social Norms

Results Wave 1



All six norm beliefs were underestimated

Correcting Misperceived Social Norms

Results Wave 2

Table 2: Treatment effects on climate donations and posterior beliefs

	(1) Donation ($)	(2) Behavior belief (post.)	(3) Norms belief (post.)
Behavior treatment	11.725**	0.279***	0.235***
	(4.675)	(0.030)	(0.030)
Norms treatment	15.674***	0.370***	0.350***
	(4.701)	(0.031)	(0.030)
N	5,991	5,988	5,976
Control group mean	249.31	0	0
z-scored	No	Yes	Yes
Controls	Yes	Yes	Yes

Statistically significant treatment effects on Donations, posterior Behaviour, and Norms belief (Causal proof of concept)

Treatment Effect Heterogeneity

Results Wave 2

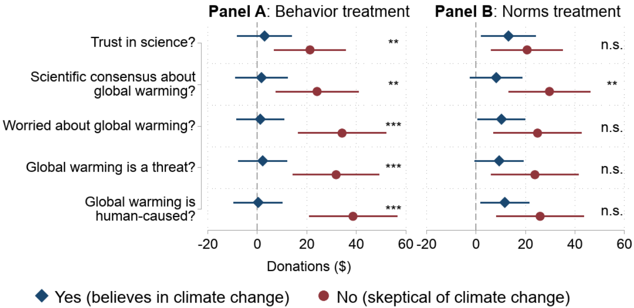
Table 3: Treatment effect heterogeneity: Prior above/below actual share

	Dependent variable: Donation ($)			
	Prior $<$ actual share		Prior $\geq$ actual share	
	(1)	(2)	(3)	(4)
Behavior treatment	14.931**		5.231	
	(5.875)		(7.701)	
Norms treatment		19.111***		4.747
		(5.387)		(9.623)
N	2,579	3,054	1,399	946
Control group mean	243.09	241.67	260.69	273.71
Controls	Yes	Yes	Yes	Yes

Positive treatment effects that we document for the full sample are almost entirely driven by those individuals whose priors are below the actual shares. Mechanism confirms belief-updating

Treatment Effect Heterogeneity

Figure 3: Treatment effect heterogeneity by climate change skepticism



Belief-updating has the potential to reduce polarisation.

Backup: Treatment Effects on Policy Support and Political Activism

Table 4: Treatment effects on support for policies and actions to fight global warming

	(1) Policies	(2) Activism	(3) All
Behavior treatment	0.088***	0.039	0.061**
	(0.026)	(0.027)	(0.026)
Norms treatment	0.066**	0.012	0.034
	(0.026)	(0.027)	(0.026)
N	5,999	5,994	5,993
z-scored	Yes	Yes	Yes
Controls	Yes	Yes	Yes

Conclusion & Implications

Conclusion

Norms matter → They're misperceived → Correcting them works → Especially for skeptics → Reducing polarization

- Behavioural Determinants: Social norms and economic preferences predict Global Warming behaviour
- Misperceptions of Social Norms: Americans systematically underestimate the share of Americans trying to fight global warming

Treatment Effects

Statistically significant treatment effects on Donations, posterior Behaviour, and Norms belief.
Many people act not out of conviction, but because they mistakenly believe that others are doing nothing.

Backup: Impact

Article | Open access | Published: 09 February 2024

Globally representative evidence on the actual and perceived support for climate action

Peter Andre, Teodora Boneva, Felix Chopra & Armin Falk

Nature Climate Change **14**, 253–259 (2024) | Cite this article

180k Accesses | **277** Citations | **2841** Altmetric | Metrics

- Andre et al. (2024b) provides causal evidence that correcting misperceived norms changes behaviour.
- The Nature paper by Andre et al. (2024a) shows that this problem exists globally and is therefore highly relevant for scalable policy.
- It received a considerable amount of attention both within and outside academia.

Backup: Key Problem of the RCT

Randomised Controlled Trial (RCT): Random assignment to treatment and control groups to identify causal effects through randomisation.

- Spillover effects
- John Henry effects (control group tries to keep up)
- **Hawthorne effects & Social Desirability Bias:** Participants might report an inflated tendency to behave sustainably, e.g. due to a desire to appear good, which may result in an artificial gap between beliefs and behaviours/norms
- **Experimenter demand effects:** obfuscating the purpose of the information provision and incentivising the donation outcome

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